Ignacio Duarte

ignacioduartef@gmail.com | (915) 234-8199 | nacho-du.github.io/portfolio | in/ignacioduartef

Education

University of Texas at El Paso, BS in Mechanical Engineering — Anticipated Fall 26

- GPA: 3.45
- **Coursework:** Mech System Control, Computational Applications in Vision and Robotics, System Dynamics, Thermal Fluid Sciences, Additive Manufacturing, Mechanics of Materials, Mech Design
- **Awards & Affiliations** Dean's List (5 Semester), UTEP SHPE/MAES, Themed Entertainment Association

Experience

Operations Intern - Bio Digester Systems, — Summer 2025

Franco & Associates - Juarez, Mex - El Paso, TX

- Operated anaerobic biodigester systems, including pump, agitation, and blower equipment to optimize biogas production
- Collected and analyzed gas composition data to ensure compliance with quality and safety standards
- Assisted with fuel distribution routing biogas to on-site system and troubleshooting combustion issues
- Coordinated and logged organic waste deliveries while ensuring safe unloading and handling procedures

Leadership

Operations Chair, – Engineering Student Leadership Council — May 2025 – May 2026

UTEP College Of Engineering

- Promoted the professional & academic development of over 5000 Students in the College of Engineering
- Facilitated and sought $50,000 in funding for 50 engineering organizations

Projects

University Rover Challenge – UTEP Capstone Projects — Jan 2026 – present

Project Manager

- Designed robotic arm manipulator with modular attachment system
- Fabricated structural components via FDM 3D printing, iterating on design for strength and weight
- Coordinated mechanical, software, and electronics sub-teams on timelines and deliverables

Sun Tracker — Two-Axis Solar Tracking System — Fall 2025

Two-axis sunflower-inspired solar tracker with closed-loop PID control

- Designed and implemented a two-DOF tracking system using four LDRs in a cross configuration to detect light direction
- Developed PID controller on Arduino Uno to drive pan and tilt SG-90 servo motors based on sensor error signals
- Fabricated custom mechanical components (base, stem, flower head, gear assemblies) using FDM 3D printing and Fusion 360

Technical Skills

Software & Tools: SolidWorks, Fusion 360, Python, Matlab (Robotics Tool Box), Blender, Unreal, Git/GitHub, Microsoft Office Suite, Latex

Technologies & Skills: CAD Modeling, CNC Lathe/Mill, FDM/SLA Printing, FEA Stress Analysis, Arduino, PID Control, Inverse/Forward Kinematics, Trajectory Planning

Languages: Native fluency In Spanish & English